

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

MODERN MANUFACTURING METHODS

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) Classify modern machining processes on the basis of the type of energy employed. | 2M |
| (b) Recommend specific advanced machining processes for cutting a glass plate. | 2M |
| (c) Define duty factor and ignition delay in Electric Discharge Machining (EDM). | 2M |
| (d) Explain the working principle of wire EDM. | 2M |
| (e) What are the limitations of chemical machining? | 2M |
| (f) Differentiate between electroplating and Electro Chemical Machining (ECM). | 2M |
| (g) List the Gases used in the plasma arc machining (PAM). | 2M |
| (h) Why deflection coil provided for EBM? | 2M |
| (i) Define prototype. What is its need? | 2M |
| (j) Write about STL file formats. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Draw a schematic diagram of Abrasive Jet Machining (AJM) system, label the various components in it and explain the working principle of AJM process. 10M

OR

- 3 Discuss the effect of amplitude of vibration, frequency of vibration, grain size, and percentage of abrasive concentration, on material removal rate (MRR) in Ultrasonic Machining System (USM). 10M

- 4 (a) Write a note on the resolidification of molten metal in EDM. 5M
(b) Illustrate the effect of magnitude of current, and frequency, on the shape and size of the craters formed during EDM. 5M

OR

- 5 (a) Explain the distribution of spark energy evolved during EDM. 6M
(b) During calculation of MRR in EDM, supply voltage was used as 60V in place of the actual supply voltage of 40 V. What is the ratio of actual and calculated MRR? Assume that the condition for maximum power delivery to the discharging circuit is satisfied. 4M

- 6 Explain ECM process with a neat sketch and discuss influence of process parameters in machining output. 10M

OR

- 7 (a) Why maskants are required in chemical machining process? Explain. 5M
(b) In what ways the electrochemical grinding differs from ordinary grinding process. 5M

Contd. in Page 2

- 8 (a) What is non-transferred and transferred mode of plasma arc? Explain their mechanism of material removal. 5M
- (b) Draw and explain components in the Electron Beam Machining (EBM). 5M

OR

- 9 What is laser? Explain how it is used to machine the materials. Enumerate the advantages, limitations and applications of laser beam machining. 10M
- 10 (a) With a neat sketch explain the working of Stereo lithography Apparatus. 5M
- (b) What are the process parameters that influence FDM process? 5M

OR

- 11 What is Laminated object manufacturing? Explain the working principle and write its advantages and disadvantages. 10M

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PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) What are the process parameters of ultrasonic machining process? | 2M |
| (b) What is Abrasive Water Jet Machining? | 2M |
| (c) What is Electrode? Give its application. | 2M |
| (d) Give the applications of Di-electric fluid. | 2M |
| (e) Applications of Electro Chemical Honing. | 2M |
| (f) What are various Techniques of Applying Maskants? | 2M |
| (g) What is Machining, give its applications? | 2M |
| (h) What are the sources of Thermal energy in manufacturing? | 2M |
| (i) Give applications of Sterolithography (SLA). | 2M |
| (j) Give any application of Laminated Object Manufacturing. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the need and characteristic features of non-traditional machining? What are the advantages of non-traditional machining process? 10M
- OR**
- 3 What is Jet machining process? Explain any two Mechanical energy-based Jet machining processes. 10M
- 4 What are the important parameters that control the material removal rate in electrical discharge machining? Briefly explain any two factors. 10M
- OR**
- 5 What is Wire cut EDM? Explain Working Principle and Applications of wire cut EDM. 10M
- 6 Difference between Chemical Machining and Electro Chemical Machining. Give their application individually. 10M
- OR**
- 7 Differentiate between Electro Chemical Grinding and Electro Chemical Honing. What are their advantages and limitations? 10M
- 8 How is the metal removal rate controlled in plasma arc machining process? Give a brief note on scope of the application and the process limitations of plasma arc machining process. 10M
- OR**
- 9 (a) What is the working principle of Electron Beam Machining? 5M
(b) Give applications, merits and demerits of electron beam machining of process. 5M
- 10 What is Additive manufacturing? Explain classifications of Additive Manufacturing Process? 10M
- OR**
- 11 Give brief explanation on Selective Laser Sintering and Fused Deposition modelling with applications? 10M
